

Reviewer Pack — Schur Positivity of Clause-Indicator Polynomials Bounds ACC^0 ...

Entry 413870e28527 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Schur Positivity of Clause-Indicator Polynomials Bounds ACC^0 Circuit Size

Entry ID: 413870e28527 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-19 23:38:23 UTC

1. Conjecture

Field A (mathematical branch): Symmetric Functions Theory **Field B** (complexity object): ACC^0 Circuit Size

Statement:

For any CNF formula with n variables, the number of distinct Schur functions that vanish on its clause-indicator polynomial is $\Theta(\log n)$ if and only if the ACC^0 circuit size is bounded by $O(\text{polylog } n)$.

Rationale (proposer's reasoning):

Schur functions encode combinatorial symmetries of polynomials, and their vanishing conditions may reveal structural constraints on clause interactions. By linking Schur positivity to circuit complexity, we explore algebraic invariants that could separate ACC^0 from NEXP.

Taxonomy category: ACC_LB_via_WILLIAMS (status at proposal time: partially_alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 4a3721f106c0aeab

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
ALGEBRIZATION	SAFE	0.85	UNCERTAIN	SAFE
KARP_LIPTON	SAFE	0.00	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_cnf(n):
        clauses = []
        for _ in range(2**n):
            clause = [random.choice([-1, 1]) * (i + 1) for i in range(n)]
            if any(clause[i] == -clause[j] for i in range(n) for j in range(i+1, n)):
                clauses.append(tuple(sorted(clause)))
        return set(clauses)

    def clause_indicator_polynomial(cnf):
        n = len(next(iter(cnf), (0,)))[0])
        poly = [0] * (2**n)
        for clause in cnf:
            sign = 1
            for var in clause:
                if var < 0:
                    sign *= -1
                var = -var
            poly[1 << (var - 1)] += sign
        return poly

    def schur_functions(n):
        schurs = []
        for i in range(2**n):
            schur = [0] * (2**n)
            schur[i] = 1
            schurs.append(schur)
        return schurs

    def count_vanishing_schurs(poly, schurs):
        count = 0
        for schur in schurs:
            product = 1
            for i in range(len(poly)):
```

```

        if poly[i]:
            product *= schur[i]
        if product == 0:
            count += 1
    return count

n = random.randint(5, 40)
cnf = generate_cnf(n)
poly = clause_indicator_polynomial(cnf)
schurs = schur_functions(n)

vanishing_count = count_vanishing_schurs(poly, schurs)

return {
    "metric_name": "vanishing_schurs",
    "metric_value": vanishing_count,
    "instances_tested": 1,
    "conjecture_holds": False,
    "counterexample": "mapping_undefined"
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or list(range(2, 30)) # Default to first 30 primes

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_value = math.sqrt(sum((result["metric_value"] - mean_value)**2 for result in results) / len(results))
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next(seed for seed, result in enumerate(results) if not result["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with detailed logging to diagnose crash cause

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	136707
2	propose	ollama_remote	qwen3:8b	0	0	171169
3	preregistration	ollama_remote	qwen3:8b	0	0	31704
4	novelty	ollama_remote	qwen3:8b	0	0	27272
5	novelty	ollama_remote	qwen3:8b	0	0	18081
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12789
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8972
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10159
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8664
10	judge	ollama_remote	qwen3:8b	0	0	19693

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 445209 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in research/audit/413870e28527.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/413870e28527.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/413870e28527.tar.gz (if generated)